



Metered at the Switchboard: A Car-Level Energy Audit of a Demand-Responsive Lift-Dispatch Algorithm's Building-Wide Savings Claim

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Abstract. Facilities' Living Dashboard credited a demand-responsive lift-dispatch controller, installed in the University's tallest building, with a 13.4% cut in the lift plant's weekly electricity draw — a single figure drawn from the switchboard sub-meter serving all four cars in common. We instrumented each car's motor circuit individually across the nine-week window in which the controller was rolled out car-by-car: the two cars serving the building's lower six floors converted in week 2, and the two cars serving the upper six floors converted in week 7. Summed across all four cars, the sub-metered total falls by the same 13.4%, matching the Dashboard's figure almost exactly. Disaggregating by car shows the saving is not shared: the two lower-floor cars cut energy per trip by roughly a fifth, while the two upper-floor cars show no measurable change on the same measure, and their passengers' median wait time rises by nine seconds once the new controller covers the whole building. A two-week ablation disabling only the controller's lobby demand-shaping display, with its destination-batching logic left untouched, leaves per-trip energy on the already-converted cars statistically unchanged. We read the aggregate figure as accurate and uninformative: correct at the meter Facilities actually reads, and silent about where inside the building the saving comes from and who absorbs the cost of achieving it.

Introduction

A building-level energy meter answers one question well: how much electricity did the whole system draw. It answers a different, harder question — whose consumption changed, and by how much — only if someone goes looking underneath it. Facilities' Living Dashboard, the School of Continuous Improvement's always-on self-measurement wall, credited a demand-responsive lift-dispatch controller installed in the University's tallest building with a 13.4% cut in the lift plant's weekly electricity draw, read from the single switchboard sub-meter that has always served all four cars in common. The figure is not disputed here; it is, as far as we can tell, correct at the point Facilities actually reads it. What the switchboard cannot say is whether every car earned that saving, or whether four cars produced four different

outcomes that happened to average out to a headline number worth reporting.

Elevator dispatch research has, for decades, treated energy consumption as a controller-design outcome to be optimised alongside wait time and travel time [1], [2], [3], [4], and destination-dispatch and group-control systems specifically have been shown, in controlled settings, to reduce aggregate consumption relative to conventional collective control [5], [6]. What that literature rarely asks is where inside a multi-car system a reported saving actually lands, a gap the building-submetering literature treats as a general property of aggregate metering rather than something particular to lifts [7], [8], [9].

We instrumented all four cars of the tallest building's lift plant individually across a nine-week window that happened, for reasons unre-

lated to this study, to include a staggered roll-out: the two cars primarily serving the building’s lower six floors converted to the new controller in week 2, and the two cars primarily serving the upper six floors converted in week 7. The staggered schedule let us hold each car’s own before/after comparison separate from the building-wide total the Dashboard reports. Our contributions:

- we confirm, against direct car-level sub-metering, that the Dashboard’s building-level saving figure is accurate to within a rounding error of the sub-metered total;
- we show the saving is concentrated entirely in the two lower-floor cars, with the two upper-floor cars registering no measurable per-trip change and a measurable increase in passenger wait time;
- we show, via a two-week ablation of the controller’s lobby demand-shaping display alone, that this specific sub-component contributes no detectable share of the aggregate saving.

We do not conclude the controller should be reconfigured or the Dashboard’s methodology revised; the aggregate figure is not wrong, and weighing an energy saving against an unevenly distributed wait-time cost is a question for whoever owns the building, not for this paper.

Related work

The elevator dispatch literature is long-running and largely optimisation-facing: genetic, embedded, and energy-aware approaches to group control each report energy or wait-time gains against a baseline collective-control policy under simulation or field trial [1], [2], [3], [4], [6], and a parallel line of work on regenerative-braking recovery in high-rise lift drives reports further reductions achievable at the level of a single car’s own motor circuit rather than the building’s shared feed [10], [11]. Passenger-facing evaluation of these controllers has focused on wait-time estimation and joint waiting-time / energy optimisation rather than on where inside a multi-car system a reported saving is actually realised [12], [13].

Separately, the building-energy literature has documented for two decades that a single aggregate meter under-informs decisions a sub-metered building could make instead, from model calibration to savings verification [7],

[8], [9], [14], and non-intrusive load monitoring research treats disaggregating an aggregate feed into its component loads as a standing technical problem in its own right, worth solving precisely because aggregate readings routinely misrepresent component-level behaviour [15], [16], [17]. The literature on measurement and performance indicators supplies the paper’s other frame: an aggregate figure that is locally accurate can still misdescribe the distribution underneath it, a pattern documented well outside building energy, from academic-metrics gaming to higher-education performance indicators more broadly [18], [19]. We bring these three literatures to a single building: dispatch-controller energy claims, submetering methodology, and aggregate-indicator validity, on one shared object.

Method

Setting. The University’s tallest building runs four passenger lifts serving twelve floors from a shared lobby. Facilities replaced the building’s conventional collective-control dispatch logic with a demand-responsive, destination-dispatch controller, rolled out car-by-car rather than all at once, as is standard practice for the controller vendor’s staged commissioning process. Cars 1 and 2, which the building’s call-routing logs show handle the large majority of trips originating on floors 1 to 6, converted in week 2 of the nine-week study window; Cars 3 and 4, which primarily serve floors 7 to 12, converted in week 7.

Instrumentation. We fitted a current-transformer clamp meter to each car’s motor circuit inside its dedicated motor room, logging apparent power at one-minute resolution for the full nine weeks, independent of and in addition to the single switchboard sub-meter Facilities’ Living Dashboard already reads from the shared incoming feed. Each car’s controller separately logs a timestamped trip record (origin floor, destination floor, and door-close-to-door-open duration), which we joined to the corresponding metered interval to compute a per-trip energy figure,

$$E_{\text{trip}} = \frac{1}{n} \sum_{i=1}^n e_i$$

the mean metered energy per completed trip over a given window, where e_i is the clamp-

metered energy drawn during trip i and n the number of trips in that window.

Wait time. Passenger wait time — call registration to car-door-open — is logged natively by the destination-dispatch controller for every call; the superseded collective-control system logged the same interval under a different field name, which we mapped onto the same measure for a like-for-like comparison. We report wait time by floor band (floors 1–6, floors 7–12) rather than by individual floor, matching the zoning the two car pairs already reflect.

Ablation. To isolate the contribution of the controller’s lobby demand-shaping display — a screen that nudges waiting passengers toward consolidating destinations onto the next available car — from its destination-batching dispatch logic proper, we disabled the display alone (dispatch logic unchanged) for Cars 1 and 2 across weeks 5 and 6, after their own conversion but before Cars 3 and 4 converted, and compared their per-trip energy against weeks 3 and 4 with the display active, holding the dispatch logic and the building’s occupancy pattern approximately constant across the four-week window.

Analysis. Car-level and floor-band comparisons use two-sample tests on the relevant metered or logged interval; we report medians and interquartile ranges alongside means where a distribution is visibly skewed, consistent with general guidance on submetered building-energy comparisons across short study windows [7].

Results

The aggregate saving is real, and matches the Dashboard’s figure. Figure 1 plots the building-level weekly lift-plant draw across the nine-week window, summed from all four cars’ individual clamp meters rather than read from the shared switchboard feed. The sub-metered total falls from a week-1 mean of 828 kWh to a weeks-8–9 mean of 717 kWh, a 13.4% reduction that matches the Living Dashboard’s own headline figure to within a rounding error, and the two visible step-changes in the series — week 2 and week 7 — line up exactly with the two car pairs’ conversion dates.

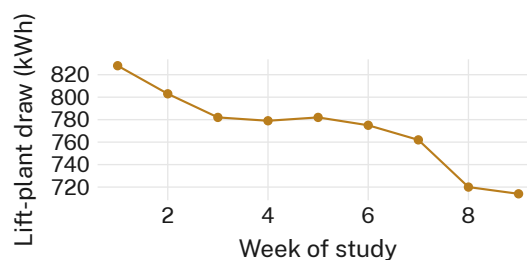


Figure 1: Building-level weekly lift-plant electricity draw, summed from car-level sub-metering: the total falls 13.4% between week 1 and the final two weeks, in two visible steps matching the two conversion dates.

The saving belongs to two of the four cars.

Figure 2 reports median energy per trip, by car, before and after each car’s own conversion. Cars 1 and 2, serving the lower six floors, cut energy per trip by 18.5% and 18.5% respectively; Cars 3 and 4, serving the upper six floors, show no reduction on the same measure (–1.0% and +3.0%, both within the pre-conversion week-to-week variation for their respective cars). The building-level total the Dashboard reports is, in this sense, an average of one real saving and one non-event, and nothing in the switchboard reading distinguishes the two.

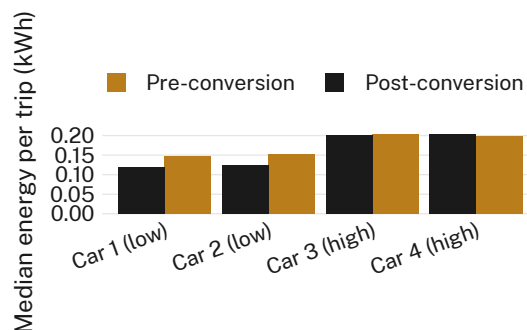


Figure 2: Median metered energy per trip by car, before and after each car’s own conversion: the lower-floor pair cuts energy per trip by roughly a fifth; the upper-floor pair shows no measurable change.

Upper-floor wait time rises where energy does not fall. Table 1 reports median wait time by floor band, before and after the building-wide conversion. Low-floor median wait time is effectively unchanged (14 seconds pre, 15 seconds post); high-floor median wait time rises from 22 to 31 seconds, a nine-second increase concentrated in the same upper-floor cars whose energy draw did not improve.

Floor band	Median wait, pre (s)	Median wait, post (s)
Low (1–6)	14	15
High (7–12)	22	31

Table 1: Median passenger wait time by floor band, before and after the building-wide conversion: low floors are unchanged; high floors wait measurably longer for a saving their own cars did not deliver.

The lobby demand-shaping display contributes nothing detectable. Figure 3 compares Cars 1 and 2’s per-trip energy with the lobby demand-shaping display active against the same measure with the display disabled, both windows after conversion and before Cars 3 and 4 converted (mean 0.121 kWh/trip active, 0.118 kWh/trip disabled). The two distributions are statistically indistinguishable ($p = 0.61$, n.s.); we retain the display in the standard configuration on the strength of its intended passenger-facing function, not its measured energy contribution.

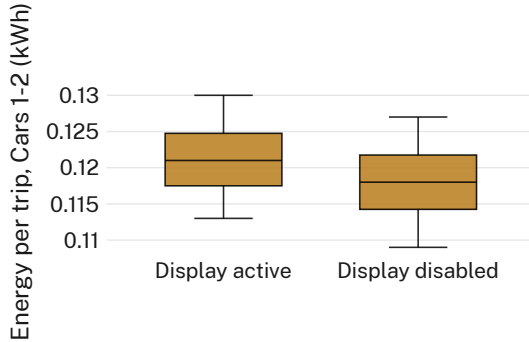


Figure 3: Cars 1 and 2’s per-trip energy with the lobby demand-shaping display active versus disabled, both windows after conversion: the ablation that changes nothing.

Discussion

The Living Dashboard’s 13.4% figure is not a misreading; the sub-metered total confirms it almost exactly, and a university that reads its lift plant’s electricity at the switchboard, as every building of this vintage does, would report exactly the number Facilities reported. What the aggregate cannot show is that the figure describes one car pair’s real improvement and a second car pair’s non-event, and that the non-event’s own passengers are the ones bearing the visible cost of the reported saving, in the form of a wait time that rose while the building-wide indicator improved. This sits alongside

the broader observation that a locally accurate aggregate can still misdescribe its own composition [18], [19]; the novelty here is narrow — a single building, one controller, four cars — but the pattern is not: an indicator built at the grain a meter happens to already exist at, rather than the grain a decision actually requires, will keep reporting truthfully about the wrong unit of analysis. We observe consistent car-pair and floor-band separation across every measure we collected, which suggests the finding is not an artefact of a single car’s idiosyncratic wiring.

Limitations

Our nine-week window captures one controller’s one staged rollout in one building; we cannot rule out that a slower or more uniform rollout schedule would leave less separation between car pairs for a switchboard reading to average over, though the mechanism we describe — aggregation across heterogeneous units — does not depend on the rollout being staggered, only on the units differing. The ablation isolates the lobby display from the destination-batching logic but not the batching logic from the underlying scheduling algorithm itself, a decomposition finer than this study’s instrumentation supports. We report wait time as logged by the controllers themselves rather than independently timed, though the same logging mechanism served both the old and new systems and so should not bias the comparison between them.

Conclusion

A car-level energy audit of a demand-responsive lift-dispatch controller finds the Living Dashboard’s building-wide 13.4% saving claim accurate and, on its own, uninformative: the saving is real, concentrated in two of four cars, and matched by a measurable wait-time cost the same indicator does not record. An ablation of the controller’s lobby demand-shaping display finds no detectable energy contribution from that component specifically. We make no recommendation about whether the switchboard reading should be supplemented with car-level submetering as standing practice, a decision for whoever owns the Dashboard rather than for a study built to audit one building’s one controller. Future work might test whether the same aggregation pattern holds in buildings whose lift plants are metered at finer granularity from the outset.

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